

PROJECT WORK

- ☐ Title
 - ☐ Project title
 - ☐ Names and IDs of students in your project group
 - ☐ Jessica Saucedo Mancias
 - ☐ Nam Bui
 - ☐ Tianyi Xu
 - ☒ Zion Swinburne 300560705
 - ☐ Date of submission
 - ☒ 03 September 2026
- ☐ Background and Data [1-3 pages]
 - ☐ State which dataset(s) your group worked on, and their source
 - ☐ Primary dataset:
 - ☐ Te Whatu Ora - Health New Zealand Pharmaceutical Data (2021-2025)
 - ☐ Health New Zealand. 2026. Pharmaceutical Data web tool version 07 August 2026 (data extracted from the Pharmaceutical Collection on 09 July 2026). URL: <https://tewhatuora.shinyapps.io/pharmaceutical-data-web-tool/> (Accessed [INSERT DATE]).
 - ☐ Totals by topic for individuals, (RC, TALB, UR, SA3, SA2, Ward, Health), 2013, 2018, and 2023 Censuses
 - ☐ From Aotearoa Data Explorer <https://explore.data.stats.govt.nz/>, 2023 Census, Totals By Topic
 - ☐ Explain briefly why the dataset is of interest, or what questions it could be used to answer; assume that the reader has never heard of your dataset
 - ☐ State the types of data in the dataset(s) and the structure of the dataset(s). Are the data numerical, categorical, or both? Time series? Coordinates? Diagnostic categories? This does NOT need to be an exhaustive list of every variable, just a few comments on the overall types.
 - ☐ Pharmac Data
 - ☐ TG2ID
 - ☐ TherapeuticGrp2
 - ☐ District
 - ☐ YearDisp
 - ☐ NumDisp
 - ☐ NumPpl
 - ☐ NHIComp
 - ☐ Census Data (Subset Selected)
 - ☐ Census night population count

- ☐ Age (life cycle groups)
 - ☐ Under 15 years
 - ☐ 15-29 years
 - ☐ 30-65 years
 - ☐ 65 years and over
- ☐ Ethnicity
 - ☐ European
 - ☐ Maori
 - ☐ Pacific Peoples
 - ☐ Asian
 - ☐ Middle Eastern/Latin American/African
 - ☐ Other ethnicity
- ☐ Maori descent indicator
 - ☐ Maori Descent
 - ☐ No Maori Descent
- ☐ Gender
 - ☐ Male
 - ☐ Female
 - ☐ Another gender
- ☐ Average number of children born
- ☐ Smoking status
 - ☐ Regular smoker
 - ☐ Ex-smoker
 - ☐ Never smoked regularly
- ☐ Disability indicator
 - ☐ Not disabled
 - ☐ Disabled
- ☐ Median total personal income
- ☐ Employment status
 - ☐ Employed Full-time
 - ☐ Employed Part-time
 - ☐ Unemployed
 - ☐ Not in the Labour force
- ☐ Average hours worked in employment per week
- ☐ State how complete the dataset(s) are (i.e. how many missing, any structure to the missing data, whether there are errors in the data)
 - ☐ Removed any values of <6 in NumPpl and NumDisp and converted to numeric
 - ☐ Check structure of these <6 values before removing from the dataset

- ☐ If you used more than one dataset, state what steps you had to take to integrate the datasets
 - ☐ Census data:
 - ☐ Select a subset of census data variables
 - ☐ Combine pharmac capital and coast with hutt valley to form Capital, Coast and Hutt Valley (due to merger)
 - ☐ Census data in wide format
 - ☐ Renamed area values to match pharmac data (census uses macrons, pharmac does not)
- ☐ Ethics, Privacy and Security [1-2 pages]
 - ☐ Brief discussion of any ethical considerations that apply to your project
 - ☐ Brief discussion of any privacy concerns that might arise connected to your project
 - ☐ Brief discussion of what steps you could take to keep your project data and results secure (you do NOT need to carry this out, you just need to talk about it in the report). Make it clear what steps you actually did take vs. any you would take if given additional time and resources.
 - ☐ Pharmac data is from official government sources, already anonymised and not identifiable to any person or specific group of persons.
 - ☐ Including census data, while you could be more specific (Statistical Area 1 groupings each have an ideal range of 100-200 households, with a maximum of 500). SA1s have a unique and publicly available identifying number which can narrow information down to a single residential block.
- ☐ EDA [3-6 pages]

For this section, do NOT try to summarize everything you can find in the dataset(s). Select a subset, highlighting features that you thought were interesting in the data. The plots do not have to be complicated; simple bar charts and scatter plots are fine.

 - ☐ Several summary tables and/or plots, each describing one, two or three variables in the data that you thought were interesting
 - ☐ Explain the definitions of the variables in each table/plot
 - ☐ Comment on the main features of each plot
 - ☐ Include suitable labels and keys for each plot
 - ☐ Make sure all plots would be readable if printed in black & white, and adjust the point sizes and/or line thicknesses to improve readability
 - ☐ Lay out all tables so that they are clearly readable and clearly labelled, and do not use excessive significant figures
 - ☐ All figures and tables should be numbered and should have captions
 - ☐ Inspect Data
 - ☐ Missing Values
 - ☐ Explore Data
 - ☐ Transformations

- ☐ The census data comes in wide format... I wanna shave down what the dataset actually includes lest I go mad
- ☐ Maybe a brief discussion paragraph (summary) on what looks interesting, and what could be investigated further?
- ☐ Individual Contributions [1 page]
 - ☐ State what contribution each member of the group made to the data preparation, the analysis and the report
 - ☐ Jessica: bibliography manager
 - ☐ Nam: repo manager
 - ☐ Tianyi: document preparation
 - ☐ Zion: data collection, census data collection
- ☐ Project Requirements:
 - ☒ ~~Select a subset of pharmaceutical products to analyse and obtain the relevant data~~
 - ☐ Plot geographical data and/or analyse time series data
 - ☐ Carry out an exploratory analysis, choosing a subset of variables and features to report
 - ☐ Determine which datasets need to be integrated to answer specific questions and/or make specific predictions
 - ☐ Carry out the analyses required to answer specific questions, and display the chosen model(s) and results
 - ☐ Consider any ethical aspects when reporting the results

About the pharmaceutical collection

The pharmaceutical collection contains claim and payment information from pharmacists for subsidised dispensings.

- The Pharmaceutical Data web tool presents information on pharmacy prescriptions for individuals. The Pharmaceutical Data web tool can be used to search, view and download summary data about initial dispensings and dispensings that were dispensed in the community and funded by the New Zealand Government. The data presented is sourced from Health New Zealand - Te Whatu Ora's Pharmaceutical Collection.

Using the collection

Start date

The collection started in 1 July 1992.

Records from before 1996 have been archived, but are available on request.

Guide for use

The pharmaceutical collection has gone through 2 major changes since 1992.

Repeat prescriptions were introduced in 1996.

The major drug key changed from Medicode, allocated by Sector Services, to Pharmacode (pharmacy stock code) in 1998.

Other small structural changes are made to the collection each year.

All money values are assumed to include GST, unless otherwise stated.

Identification of co-prescribing, which is the analysis of drugs being prescribed and taken together, has been a problem with historical data.

Collection methods — guide for providers

The main source of the data is Sector Operations' GTPS. Extra data is provided by:

- Stats NZ geographical and census data — claimant data is geocoded by the national collections and reporting geocoding engine
- Sector Operations — provider reference data
- Pharmac — the Pharmac Schedule and mappings.

Frequency of updates

Health NZ gets monthly extracts from Sector Operations and Pharmac.

Security of data

The collection is accessed by authorised Health NZ and Pharmac staff for the purposes of:

- maintenance
- data quality
- audit
- analytics.

Authorised members of the Ministry of Health and Health NZ have access to the collection for analytical purposes, using Business Objects and the secure Health Information Network. Business Objects contains a subset of the data described in the Data Dictionary.

Access to the collection requires approval from Health NZ.

Business Objects is being decommissioned.

Privacy issues

We are required to make sure that the release of information meets the conditions of any legislation related to the privacy of health information. In particular the:

- Official Information Act 1982
- Privacy Act 2020
- Health Information Privacy Code 2020.

Information available to the general public is of a statistical and non-identifiable nature. A data release policy covers approval where appropriate from a health ethics committee.

We are responsible for the privacy of NHI-related information. NHIs were provided for approximately 97% of dispensings in 2010.

National reports and publications

Summarised pharmaceutical data can be accessed using the Pharmaceutical data web tool and in Pharmac's annual reports.

Data provision

Customised datasets or summary reports are available on request. These can be electronic or on paper.

Our team can help to define the specifications for a request and are familiar with the strengths and weaknesses of the data.

We also offer a peer review service to make sure Health NZ data is reported appropriately when published by other organisations.

Citation

Health New Zealand. 2026. Pharmaceutical Data web tool version 07 August 2026 (data extracted from the Pharmaceutical Collection on 09 July 2026). URL:

<https://tewhatuora.shinyapps.io/pharmaceutical-data-web-tool/> (Accessed OFTEN).

Summary

```
> summary(pharms)
  Type      TG2ID      TherapeuticGrp2      District      YearDisp      NumDisps      NumPp1      NHIComp
Length:614   Min.   : 101   Length:614   Length:614   Min.   :2021   Length:614   Length:614   Min.   :100
Class :character 1st Qu.:1001 Class :character 1st Qu.:2022 Class :character 1st Qu.:100
Mode  :character Median :1419 Mode :character Median :2023 Mode :character Median :100
              Mean  :1687              Mean  :2023              Mean  :100
              3rd Qu.:2232              3rd Qu.:2024              3rd Qu.:100
              Max.   :4525              Max.   :2025              Max.   :100
```

Dispensings

```
> top10
# A tibble: 10 x 2
  TherapeuticGrp2      total_dispensings
  <chr>          <dbl>
1 Analgesics      27267705
2 Antibacterials  16778612
3 Agents Affecting the Renin-Angiotensin System 13907159
4 Lipid-Modifying Agents 12876340
5 Non-Steroidal Anti-Inflammatory Drugs 11688851
6 Antidepressants 11504069
7 Antiulcerants   10800572
8 Antithrombotic Agents 9806567
9 Diabetes        9346233
10 Vitamins       9235016
```

```
> district_total
# A tibble: 20 x 2
  District      total_dispensings
  <chr>          <dbl>
1 Counties Manukau 33283030
2 Waitemata       33207402
3 Canterbury       32020241
4 Auckland        24795754
5 Waikato          24573137
6 Southern        19350410
7 Capital & Coast 15290791
8 Bay of Plenty   14449210
9 Northland       11045409
10 MidCentral     11036632
11 Hawke's Bay    10102263
12 Nelson Marlborough 8699814
13 Hutt Valley     8467986
14 Taranaki       7096556
15 Lakes          5942383
16 Whanganui      4197722
17 South Canterbury 3646440
18 Wairarapa      2904493
19 Tairāwhiti     2644127
20 West Coast     1945020
```

```
> highest_districts
# A tibble: 5 x 2
  District      total_dispensings
  <chr>          <dbl>
1 Counties Manukau 33283030
2 Waitemata       33207402
3 Canterbury       32020241
4 Auckland        24795754
5 Waikato          24573137
```

```
> lowest_districts
# A tibble: 5 x 2
  District      total_dispensings
  <chr>          <dbl>
1 West Coast     1945020
2 Tairāwhiti     2644127
3 Wairarapa      2904493
4 South Canterbury 3646440
5 Whanganui      4197722
```

```
> change_top10
# A tibble: 10 x 5
  TherapeuticGrp2      dispensing_2021      dispensing_2025      change      percent_change
  <chr>          <dbl>          <dbl>          <dbl>          <dbl>
1 Analgesics      4747437      5833774 1086337      22.9
2 Vitamins        1510462      2295146  784684      51.9
3 Lipid-Modifying Agents 2263553      2935628  672075      29.7
4 Diabetes        1600157      2181323  581166      36.3
5 Agents Affecting the Renin-Angiotensin System 2529538      3055503  525965      20.8
6 Antibacterials  2983307      3479202  495895      16.6
7 Non-Steroidal Anti-Inflammatory Drugs 2029315      2504596  475281      23.4
8 Antiulcerants   1987013      2338025  351012      17.7
9 Antidepressants 2150465      2466546  316081      14.7
10 Antithrombotic Agents 1880590      2054936  174346      9.27
```

```
> highest_combinations
# A tibble: 10 x 3
  District      TherapeuticGrp2      total_dispensings
  <fct>          <fct>          <dbl>
1 Counties Manukau Analgesics      3769939
2 Waitemata       Analgesics      3199072
3 Canterbury       Analgesics      3163080
4 Waikato          Analgesics      2409039
5 Counties Manukau Antibacterials  2376787
6 Auckland        Analgesics      2357512
7 Waitemata       Antibacterials  2130743
8 Southern        Analgesics      1952395
9 Canterbury       Antibacterials  1701911
10 Canterbury       Antidepressants 1693808
```

```
> lowest_combinations
# A tibble: 10 x 3
  District      TherapeuticGrp2      total_dispensings
  <fct>          <fct>          <dbl>
1 Tairāwhiti     Vitamins        49593
2 West Coast     Diabetes        56863
3 West Coast     Non-Steroidal Anti-Inflammatory Drugs 73218
4 Wairarapa      Vitamins        81359
5 West Coast     Antithrombotic Agents 81685
6 West Coast     Vitamins        82208
7 West Coast     Antibacterials  84336
8 Tairāwhiti     Antidepressants 88650
9 Wairarapa      Diabetes        91005
10 West Coast     Antidepressants 92151
```

Patients

```
> top10
# A tibble: 10 x 2
  TherapeuticGrp2      total_patients
  <chr>              <dbl>
1 Analgesics         9881842
2 Antibacterials     8974859
3 Non-Steroidal Anti-Inflammatory Drugs 7006842
4 Antihistamines     3886489
5 Antiulcerants      3747623
6 Corticosteroids Topical 3620712
7 Agents Affecting the Renin-Angiotensin System 3557699
8 Lipid-Modifying Agents 3341387
9 Antidepressants    3137667
10 Vitamins           2812239
```

```
> district_total
# A tibble: 20 x 2
  District      total_patients
  <chr>         <dbl>
1 Counties Manukau 13511432
2 Waitemata       13355848
3 Canterbury      12228961
4 Auckland        10220722
5 Waikato         9585331
6 Southern        7258096
7 Capital & Coast 5991100
8 Bay of Plenty   5622340
9 Northland       4118038
10 MidCentral     3960012
11 Hawke's Bay    3820554
12 Nelson Marlborough 3219588
13 Hutt Valley    3196221
14 Taranaki       2628079
15 Lakes          2368205
16 Whanganui      1531461
17 South Canterbury 1347509
18 Wairarapa      1069583
19 Tairāwhiti     1015919
20 West Coast     681619
```

```
> highest_districts
# A tibble: 5 x 2
  District      total_patients
  <chr>         <dbl>
1 Counties Manukau 13511432
2 Waitemata       13355848
3 Canterbury      12228961
4 Auckland        10220722
5 Waikato         9585331
```

```
> lowest_districts
# A tibble: 5 x 2
  District      total_patients
  <chr>         <dbl>
1 West Coast     681619
2 Tairāwhiti     1015919
3 Wairarapa      1069583
4 South Canterbury 1347509
5 Whanganui      1531461
```

```
> change_top10
# A tibble: 10 x 5
  TherapeuticGrp2      patients_2021 patients_2025 change percent_change
  <chr>              <dbl>         <dbl>   <dbl>      <dbl>
1 Analgesics         1749590       2048374 298784    17.1
2 Non-Steroidal Anti-Inflammatory Drugs 1237133       1479989 242856    19.6
3 Antibacterials     1617911       1852509 234598    14.5
4 Vitamins           462889        696412 233523    50.4
5 Antihistamines     691221        842319 151098    21.9
6 Lipid-Modifying Agents 595608       746469 150861    25.3
7 Agents Affecting the Renin-Angiotensin System 653538       773023 119485    18.3
8 Antiulcerants      701838       792597 90759     12.9
9 Antidepressants    599367       654226 54859      9.15
10 Corticosteroids Topical 701613       744273 42660      6.08
```

```
> highest_combinations
# A tibble: 10 x 3
  District      TherapeuticGrp2      total_patients
  <fct>         <fct>              <dbl>
1 Counties Manukau Analgesics         1479704
2 Waitemata       Analgesics         1247299
3 Counties Manukau Antibacterials     1241653
4 Waitemata       Antibacterials     1150282
5 Canterbury      Analgesics         1071136
6 Auckland        Analgesics          954308
7 Counties Manukau Non-Steroidal Anti-Inflammatory Drugs 949858
8 Canterbury      Antibacterials     931671
9 Waitemata       Non-Steroidal Anti-Inflammatory Drugs 889840
10 Waikato         Analgesics          883266
```

```
> lowest_combinations
# A tibble: 10 x 3
  District      TherapeuticGrp2      total_patients
  <fct>         <fct>              <dbl>
1 Tairāwhiti     Vitamins           13604
2 West Coast     Corticosteroids Topical 18069
3 West Coast     Antihistamines     20183
4 West Coast     Vitamins           22221
5 Wairarapa      Vitamins           23735
6 West Coast     Antidepressants    25246
7 Tairāwhiti     Antidepressants    25406
8 West Coast     Lipid-Modifying Agents 26116
9 Wairarapa      Corticosteroids Topical 28834
10 West Coast     Agents Affecting the Renin-Angiotensin System 29936
```